TITLE :

Botbrain : Smart Campus Navigation for Chanakya University

Abstract

Getting around a big campus can feel confusing, especially for newcomers. Bot brain is designed to make life easier by helping chanakya university students find their way. This intelligent assistant guides users from place to place, showing the best routes and providing building info. By using proven search techniques and a simple user interface, Bot Brain aims to save students time and help them explore their university with confidence.

Introduction

Large university campuses can be overwhelming, even for returning students. Orientation maps are helpful, but they can't answer questions or adapt to a student's needs. Bot Brain hopes to bridge this gap. With a smart, interactive system, students can get directions, learn about campus buildings, and quickly locate services—all from their phone or computer.

Problem Statement

Many students struggle to find lecture halls, administrative offices, or amenities when they're new to the campus. Paper maps and static images don't give real-time directions or explain the quickest way to get places. There's a real need for a smarter, easier way to navigate Chanakya University without getting lost or feeling frustrated.

Objectives

1. Model Chanakya University as a network of pathways between key buildings.
2. Provide multiple ways (BFS, DFS, UCS, A*) for the system to figure out the best route.
3. Build an interface where users can ask for directions and get useful information.
4. Make it easy to compare which algorithm gives the shortest or fastest path.
5. Lay a foundation for future features like visual maps or personalized suggestions.

Scope

This project will focus on getting students from point A to point B on campus using a text-based application. The first version will use search algorithms to map routes and provide basic info about main locations. Features like graphical maps or real-time alerts could be added later if time allows.

Requirements Document

- **Campus Map:** A list of all primary locations like
- academic buildings
- Library
- hostels
- canteen
- auditorium
- sports area
- Gates
- and how they connect

1. **Path Details:** Info about distance between locations, any one-way routes, and any spots students can't access.
2. **Search Algorithms:** Code for BFS, DFS, UCS, and A*, each showing step-by-step how they find routes.
3. **User Interface:** Simple text prompts to ask for a starting point, destination, and which pathfinding method to use.
4. **Building Info:** Short descriptions and details for each building—like what it's used for and when it's open.
5. **Comparison Tools:** Basic stats so users can see which path or algorithm is most efficient.

Tool/Technology List

1. **Programming Language:** Python
2. **Framework:** Flask (for web interface)
3. **Data Storage:** In-memory lists/dictionaries
4. **APIs:** Optional use of Google Maps
5. **Frontend:** Simple HTML

6. **Version Control:** Git with a remote repository
7. **Hosting:** Heroku, PythonAnywhere, or Render for putting the web app online